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Comparative analysis of ground reaction forces and spatiotemporal gait parameters in older adults with sway-back posture and chronic low back pain: a cross-sectional study

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Abstract

Background Gait alterations associated with postural deviations are a significant factor contributing to functional limitations in older adults. Among these, sway-back posture has been linked to chronic low back pain (CLBP), defined as pain persisting for more than three months. This study aimed to analyze ground reaction forces (GRFs), loading and unloading rates, spatiotemporal gait parameters, and the asymmetry index (ASI) in older adults with sway-back posture and CLBP (SBCLBP) without adjusting for walking speed.

Methods A total of 36 older adults were included and categorized into three groups: 12 with SBCLBP, 12 with CLBP without sway-back posture, and 12 without CLBP. GRFs and spatiotemporal gait parameters—including stride time, cadence, stride length, gait speed, and stance phase duration—were recorded for all participants. GRFs were analyzed at the anteroposterior peaks during heel contact (Fx1) and push-off (Fx2) phases, as well as at the vertical force peaks at heel contact (Fy1), mid-stance (Fy2), and push-off (Fy3) phases. Additionally, mediolateral force peaks (Fz) during heel contact were assessed. The ASI was calculated for all participants. Between-group differences were examined using one-way ANOVA and ANCOVA.

Results Cadence, stride length, gait speed, and Fy2 values were significantly lower in the SBCLBP and CLBP groups compared to the control group. Additionally, these parameters were significantly lower in the SBCLBP group than in the CLBP group. However, the loading rate demonstrated greater variability across participants. No significant differences in ASI were observed among the groups.

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Conclusions Older adults with SBCLBP exhibit distinct gait characteristics compared to those with CLBP alone and those without CLBP. These differences may be attributed to structural postural alterations, distinguishing this subgroup from individuals with isolated CLBP. These findings underscore the need for targeted rehabilitation strategies tailored to the specific biomechanical alterations observed in this population. Future research should focus on optimizing intervention protocols to enhance stability and mitigate pain-related gait impairments associated with aging.

Trial registration Current Controlled Trials using the UMIN Clinical Trials Registry website with ID number of, UMIN000055653 “Retrospectively registered” at 27/09/2024.

Keywords Chronic low back pain, Older adults, Gait, Ground reaction forces, Spatiotemporal, Sway-back

Introduction

Chronic low back pain (CLBP) exhibits a significant age-related increase, impacting a substantial segment of the elderly population worldwide [1, 2]. Epidemiological estimates suggest that approximately 21–68% of individuals aged 60 years or older experience CLBP annually [3]. The high prevalence of CLBP is of particular concern due to its substantial contribution to functional impairments in this age group [4]. This high prevalence is of particular concern due to its strong association with progressive functional impairments, particularly in locomotor abilities such as walking [5].

CLBP is linked to alterations in gait dynamics through several biomechanical and neuromuscular mechanisms. These include reduced trunk control, proprioceptive deficits, and muscular imbalances, all of which contribute to compensatory gait adaptations [6, 7].

Studies have shown that individuals with CLBP often exhibit decreased gait speed, shorter stride length, increased stance time, and alterations in ground reaction forces (GRFs), suggesting a compromised ability to generate and control movement [8, 9]. Additionally, deficits in spinal stability associated with CLBP can lead to increased reliance on passive structures, further exacerbating postural control difficulties [10].

Walking function is essential for maintaining independence in older adults. To objectively assess gait performance, researchers analyze various spatiotemporal parameters—such as stride time, cadence, stride length, gait speed, and stance time—as well as GRFs [11, 12]. These metrics establish a normative basis for identifying gait deviations that may reflect underlying functional impairments in the elderly [13]. GRFs, which describe the interaction forces between the ground and the body during foot contact, provide critical insights into dynamic gait mechanics [14]. Specifically, altered loading rates and asymmetrical force distribution in CLBP patients suggest adaptations aimed at minimizing discomfort but may lead to further musculoskeletal deterioration [15].

The presence of postural deviations, such as sway-back posture, may amplify these impairments. Sway-back posture is characterized by excessive posterior displacement

of the thorax relative to the pelvis, often accompanied by weakened gluteal muscles and altered sagittal balance [16]. This condition may contribute to abnormal load distribution during gait, affecting both postural stability and pain perception [17]. While some studies have examined the link between CLBP and spatiotemporal gait parameters, comprehensive analyses incorporating postural misalignments remain limited [18]. Understanding these relationships is critical, as both CLBP and postural abnormalities can independently and synergistically impact gait mechanics, increasing the risk of falls and further functional decline [19].

Recent research highlights significant differences in gait parameters between individuals with CLBP and those without, including increased stance duration, reduced stride length, and alterations in GRFs [19, 20]. However, the extent to which sway-back posture modifies these parameters remains unclear. Given the potential interaction between chronic pain and postural misalignment, investigating these factors together can provide valuable insights into their combined impact on mobility in older adults [21, 22].

Moreover, the propensity for postural shifts in CLBP patients, often due to muscular deficiencies, particularly the weakening of the gluteal muscles, fosters a sway-back posture [23]. Although multiple studies have corroborated the substantial influence of sagittal imbalances on pain and disability [24, 25], the literature documenting these postural dynamics in relation to spatiotemporal gait parameters remains limited [26].

Recent works have highlighted the biomechanical differences during tasks such as gait in individuals with CLBP compared to pain-free controls [27], while most studies have primarily focused on the ages of the population [5, 7]. However, to our knowledge, little research has been reported on the effects of posture on gait variables in elderly people with CLBP. Studies indicate that changes in the spine of elderly individuals lead to alterations in gait variables [28]. Additionally, modifications in the spine associated with sway-back posture, including an increase in the posterior tilt of the pelvis, trunk, and thoracic kyphosis, can further affect these variables,

while also increasing stress in the lumbar region [29]. It has been reported that changes in the increase of thoracic kyphosis and the decrease of spinal lordosis, along with the resulting alterations in the lower limbs [30], can contribute to the presence of imbalance in older adults and ultimately lead to an increase in chronic pain in these individuals. Recent studies have shown that changes in postural alignment and the experience of chronic pain, such as CLBP, can have a greater impact on the gait of older adults, especially when compared to individuals who experience only one of these factors. The importance of examining changes in gait in elderly individuals who experience changes in posture along with chronic pain, this can lead to effective solutions aimed at reducing the problems faced by the elderly, particularly balance issues and the prevention of increased fall risk [31]. Previous studies have shown that body posture and movement patterns can play a significant role in the exacerbation of pain. Therefore, understanding these changes in movement patterns and improving posture may contribute to reducing pain levels and enhancing the quality of life for individuals [32].

This study aims to analyze the differences in ground reaction forces (GRFs), loading and unloading rates, spatiotemporal gait parameters, and asymmetry index (ASI) among three groups of older adults: (1) those with sway-back posture and chronic low back pain (SBCLBP), (2) those with chronic low back pain (CLBP) without postural deviations, and (3) those without CLBP. The objective is to determine whether the presence of sway-back posture introduces distinct biomechanical and gait alterations beyond those observed in individuals with CLBP alone. We hypothesize that SBCLBP will exhibit greater impairments in gait parameters and GRF amplitudes compared to both CLBP and control groups, while asymmetry in force application will remain minimal across all groups.

Materials and methods

Study design

This observational cross-sectional study was conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [33]. The study protocol was approved by the Ethics Committee of the Medical Center Office of Shiraz Rehabilitation Sciences University (IR.SUMS.REHAB.REC.1399.034). Informed consent was obtained from all participants, adhering to the ethical principles of the Declaration of Helsinki.

Setting and participants

The study was carried out at a clinical research center in the rehabilitation laboratory of Shiraz University, Iran, from December 14, 2020, to March 3, 2021. Three

participant groups were recruited: older adults with SBCLBP, those with CLBP only, and those without any history of CLBP. The inclusion criteria targeted participants aged 60–85 years [5].

Participants in the CLBP group were included based on a numeric pain rating scale (NPRS) for current, minimum, and maximum pain experienced over the past 24 h, pain experienced at least 4 days per week, and a minimum pain duration of 3 months [2]. Exclusion criteria were walking with personal assistance or assistive devices, radicular pain below the knee, known spinal pathology other than osteoarthritis (e.g., recent trauma, vertebral compression fractures, ankylosing spondylitis, metastatic carcinoma), a history of lumbar surgery, severely limited mobility (e.g., requiring assistive devices for household ambulation), progressive neurological disorders, or terminal illness.

Participants with SBCLBP were included if they met the inclusion and exclusion criteria of the CLBP group and exhibited a sway-back posture. Details on sway-back posture assessment are described below. Participants in the non-CLBP group were required to have no history of LBP at enrollment and were excluded if they had undergone lumbar surgery, received treatment for CLBP in the previous 6 months, or had any of the aforementioned exclusion criteria, also lacked sway back posture and other changes in body alignment. In general, all participants who entered this study had the ability to walk independently and without the need for a companion or assistive device.

Self-Reported measures

Pain in older adults with CLBP were assessed using the Numerical Pain Rating Scale (NPRS) [34, 35]. Additional measurements included demographic variables such as age, sex, leg length, height, weight, and body mass index (BMI), was calculated by dividing the participant's weight (kg) by the square of their height (m^2) [13]. The NPRS, ranging from 0 ("painless") to 10 ("worst possible pain"), was used to evaluate current pain intensity, as well as the highest and lowest pain levels in the past 24 h. These scores were averaged to yield a composite pain intensity rating [36].

Sample size calculation

The sample size was determined based on previous research on gait parameters with similar descriptive and exploratory objectives [37]. Using G*Power software, a minimum of 12 participants per group was required to detect a statistically significant effect (partial $\eta^2 = 0.338$) with 95% confidence and adequate power (0.80), which pertains to the gait parameters. To account for potential dropouts, we aimed to include at least 15 participants in each group.

Evaluation of sway-back posture

The evaluation of sway-back posture was conducted in the first step. Participants who met the inclusion criteria for the study were classified according to clinical descriptions of various standing postures in the sagittal plane to investigate the presence of sway-back posture [38]. The participant was asked to stand in a comfortable posture, looking ahead, with the feet positioned so that the legs are parallel to each other and the toes are pointed straight ahead. Next, we attached markers to the subject's left side on the external auditory meatus, acromion, greater trochanter, and lateral malleolus using double-sided tape. The subject was photographed from the left using



Fig. 1 Sway-back posture in older adults with chronic low back pain. Note: AM—external auditory meatus; A—acromion; GT— greater Trochanter; LM—lateral malleolus

a camera placed 2.5 m away. The passage of the plumb line and the examination of the various sections were conducted to assess the posture (Fig. 1). Participants with sway back posture, characterized by a posterior displacement of the thorax relative to the pelvis (i.e., a backward trunk lean relative to the hips), a long thoracic kyphosis, and a flattened lumbar angle (i.e., less lumbar lordosis), were evaluated by clinical assessment conducted by a corrective exercise expert [39].

Spatiotemporal gait parameters

Gait data were captured using a Qualisys™ motion capture system (Qualisys, Sweden) with eight cameras, recording trajectories of reflective markers attached to the participants at 120 Hz, and Qualisys Track Manager Software (QTM). The system was calibrated according to manufacturer guidelines before each session. Kinematic data were low-pass filtered with a 10 Hz cutoff. A Kistler force platform synchronized with the Qualisys system was used to identify gait events, kinetics, and trajectory. Retro-reflective markers were positioned on participants' heels and toes for accurate spatiotemporal gait measurement during multiple 10-meter walking trials. One gait cycle was defined as the duration from initial contact of one foot to the subsequent contact of the same foot. Among the walking cycles, the cycle that started with the heel striking the force plate was selected until the same heel made contact again in the next step. Parameters assessed included cadence, stride length, gait speed, and stance time [40]. Stride length was calculated as the anterior-posterior displacement of the right heel marker during two consecutive heel strike events, and Stance phase percentage (ST/GC) was the percentage of stance phase in one gait cycle. To evaluate these parameters, the participant performed three trials with both feet, and the overall average was reported.

Ground reaction forces (GRFs)

A kistler force plate was used to record the forces applied to the leg during walking. The force plate data were collected with a frequency of 1200 Hz and filtered with a Butterworth low pass filter with a cut-off frequency of 10 Hz. The peaks of the vertical force (FY), anteroposterior force (FX), and mediolateral force (FZ) (Fig. 2) applied on average of the right, and left legs were used for the final analysis. In total, six clinical trials were conducted, with three trials for each leg, in which GRFs were measured three times for the right foot and three times for the left foot on a force plate, from the moment of heel strike to toe-off. GRFs threshold of 10 N was used for the identification of heel strike event. The GRF values were recorded along the vertical (y), anteroposterior (x), and mediolateral (z) directions as follows: the peak of the vertical GRF (FyMAX) and the indentation of the

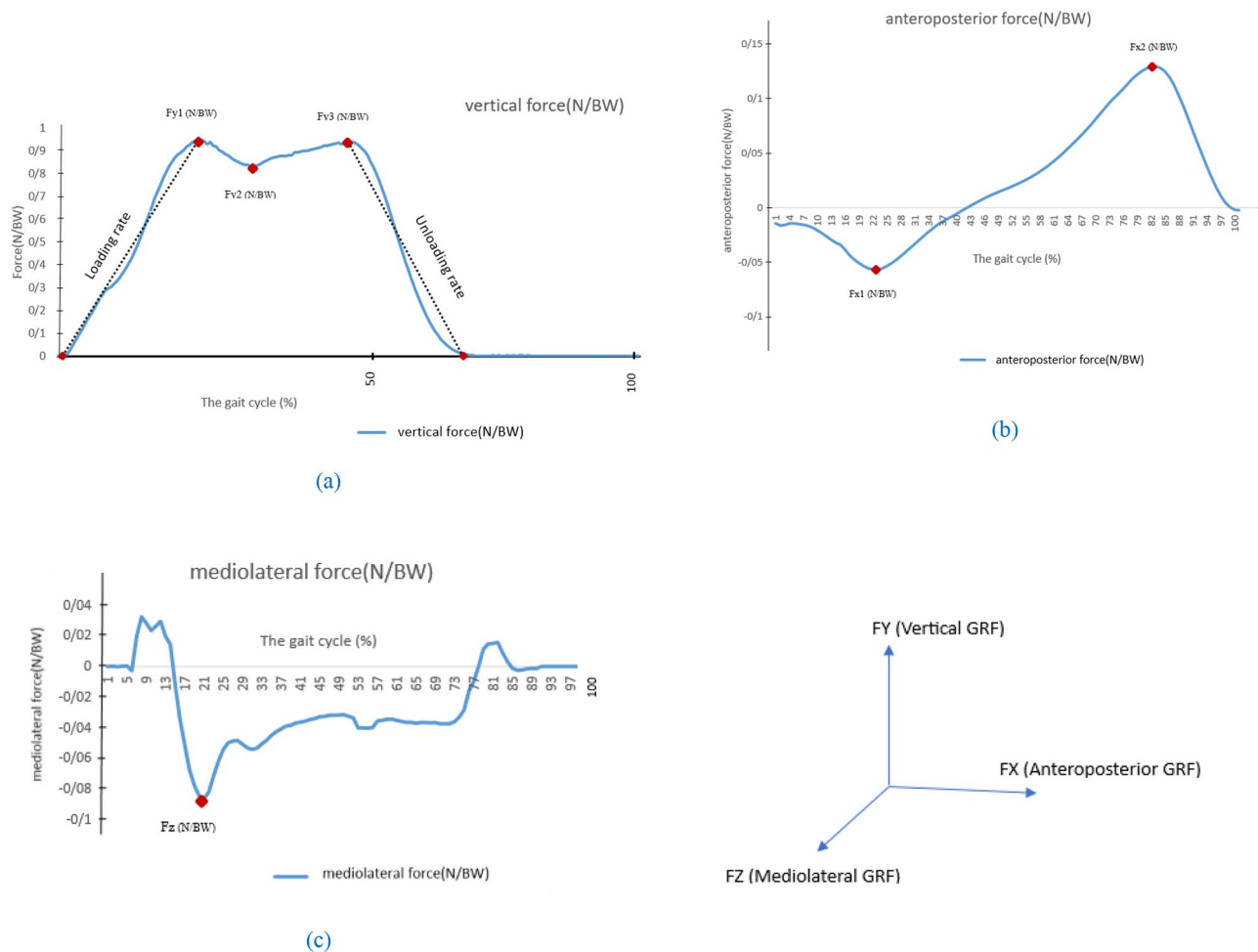


Fig. 2 The vertical (a), anteroposterior (b), and mediolateral forces (c) of SBCLBP subject. This figure provides a comprehensive overview of the ground reaction forces (GRFs) discussed throughout the article

curve (the minimum value between the two peaks of the curve); the positive (FxMAX) and negative (FxMIN) peaks of the anteroposterior curve; and the negative peak in the mediolateral direction (FzMIN). Stride length was normalized by leg length and GRF by body weight [13]. The loading and unloading rates of vertical components of ground reaction force were also evaluated in this study. It was done based on the method described by Chockalingam et al. [41] and Schizas et al. [42]. Loading rate and unloading rate are estimated as the difference in force over difference in time.

$$\text{Loading rate} = \Delta F (\text{load}) / \Delta t$$

$$\text{Unloading rate} = \Delta F (\text{unload}) / \Delta t$$

Data processing

Data were processed using the QTM software to display and identify the markers' trajectories and their six degrees of freedom using the Automatic Identification

Model. Each gait event was labelled, and foot contact and foot off gait events were labelled to identify stance and swing phases. A low-pass filter was used to remove the noise without affecting the true signals. The output was transmitted to a computer through analogue-digital converter, then to QTM and sampled at a frequency of 120 Hz. Participants' information and data obtained from walking tasks were extracted and analyzed by a biomechanics expert with 10 years of experience in the field.

Asymmetry index (ASI)

Gait symmetry was evaluated using the Asymmetry Index (ASI), as described by Herzog et al. [43]. An ASI value of zero indicates perfect symmetry, while values above or below zero denote asymmetry. The ASI was used to evaluate the symmetry of movements as well as the forces applied to the legs in the ground reaction forces. The asymmetry was measured based on the following equation:

Table 1 The characteristics of older adults with SBCLBP, CLBP, and without CLBP participants who participated in the study

Characteristics	SBCLBP Group (n = 12)	CLBP Group (n = 12)	Without CLBP Group (n = 12)	P-value
Sex, n	Male 5, Female 7	Male 6, Female 6	Male 4, Female 8	0.710
Age (y)	69.08 ± 7.28	70.08 ± 4.35	69.66 ± 1.92	0.288
Body Height (cm)	158.75 ± 8.06	163.66 ± 5.05	162.16 ± 5.7	0.173
weight (kg)	57.75 ± 11.61	59.33 ± 6.13	59.33 ± 4.90	0.875
Body mass index (kg/m ²)	25.74 ± 3.82	22.08 ± 1.25	22.52 ± 0.64	0.308
Average NPRS (0–10)	3.42 ± 1.08	3.42 ± 0.79	N/A	0.017

Abbreviation: CLBP, Chronic low back pain; SBCLBP, sway-back and chronic low back pain; NA; Not applicable, NPRS, Numerical pain rating scale. Values stands for mean ± SD

$$ASI = \frac{X_{right} - X_{left}}{(X_{right} + X_{left})/2} \times 100$$

Statistical analysis

All statistical analyses were performed using SPSS version 28 (IBM, Armonk, NY, USA). The Shapiro-Wilk test was applied to assess data normality, while Levene's test evaluated homogeneity of variances. Descriptive statistics, including demographic characteristics and mean pain intensity, were calculated for all three groups.

For normally distributed variables, one-way ANOVA was utilized to compare cadence, stride length, gait speed, loading rate, and force asymmetry (Fz1, Fz2, Fz3) among the SBCLBP, CLBP, and control groups. Additionally, ANCOVA was employed for group comparisons of force components (Fz1, Fz2, Fy2), adjusting for baseline gait speed. All statistical tests were conducted with a significance level of $\alpha = 0.05$.

Results

Demographic and clinical characteristics of participants

A total of 51 individuals were screened for eligibility, of whom 15 were excluded due to a history of lumbar surgery, the presence of radicular symptoms extending below the knee, or inconsistencies identified during postural assessment. The final sample comprised thirty-six participants, stratified into three groups: older adults with SBCLBP ($n = 12$, age range: 63–80 years, mean age: 69.8 ± 7.28 years), older adults with non-specific chronic CLBP ($n = 12$, age range: 65–80 years, mean age: 70.08 ± 4.35 years), and a control group of older adults without CLBP ($n = 12$, age range: 66–72 years, mean age: 69.66 ± 1.92 years). All groups exhibited a normal Body Mass Index (BMI), spanning from 18.5 to 24.9 kg/m². No significant demographic differences were observed across the groups, apart from the Numeric Pain Rating Scale (NPRS) scores, which were exclusively elevated in the CLBP cohorts. The characteristics of the study participants are detailed in Table 1.

Spatiotemporal gait parameters

The analysis of spatiotemporal gait parameters revealed significant intergroup differences. Notably, stride time, cadence, stride length, gait speed, and stance time all varied significantly ($p < 0.001$ for stride time, $p = 0.002$ for cadence, $p = 0.025$ for stride length, $p = 0.003$ for gait speed, $p = 0.007$ for stance time). Participants with SBCLBP demonstrated reduced cadence, stride length, and gait speed coupled with prolonged stride and stance times compared to the other groups. These parameters are summarized in Table 2.

Ground reaction forces (GRFs)

Differential ground reaction forces were apparent among the groups. The vertical force during mid-stance (second peak FY) and loading rates were significantly different, with the non-CLBP group displaying the highest second

Table 2 Between-group differences for Spatiotemporal parameters of self-selected gait for older adults with SBCLBP, CLBP, and without CLBP participants who participated in the study

Gait Parameter	SBCLBP Group (n = 12) Mean ± SD (min-max)	With CLBP (n = 12) Mean ± SD (min-max)	Without CLBP (n = 12) Mean ± SD (min-max)	P-value
Stride time (s)	1.41 ± 0.23 (1.12–1.83)	1.22 ± 0.09 (0.99–1.40)	1.07 ± 1.95 (0.74–1.31)	< 0.001
Cadence (steps/min)	86.15 ± 3.85 (77.67–94.63)	96.48 ± 1.70 (92.72–100.23)	101.08 ± 1.57 (102.12–109.04)	0.002
Stride length/ leg length	0.93 ± 0.19 (0.74–1.13)	0.97 ± 0.11 (0.74–1.16)	1.09 ± 0.55 (0.97–1.30)	0.025
Gait speed (m/s)	0.68 ± 0.25 (0.34–1.05)	0.84 ± 0.14 (0.63–1.03)	0.97 ± 0.14 (0.72–1.19)	0.003
Stance (GC)	0.66 ± 0.04 (0.62–0.72)	0.63 ± 0.03 (0.59–0.70)	0.62 ± 0.02 (0.59–0.66)	0.007

Stride length normalized to leg length. Abbreviation: CLBP, Chronic low back pain; (SBCLBP): older adults with CLBP and sway-back, (With CLBP): older adults with CLBP, (Without CLBP): older adults without CLBP. CI, Confidence Interval; GC, Gait cycle; s, second; m, meter; min, minute

Table 3 GRF (Average left and right foot) of older adults with SBCLBP, CLBP, and without CLBP participants who participated in the study

Gait parameter	SBCLBP Group (n = 12) Mean $\pm$ SD (min-max)	With CLBP Group (n = 12) Mean $\pm$ SD (min-max)	Without CLBP Group (n = 12) Mean $\pm$ SD (min-max)	P-value
Fy1 (N/BW)	0.94 $\pm$ 0.057 (0.085–0.105)	0.989 $\pm$ 0.028 (0.942–1.035)	1.014 $\pm$ 0.020 (0.978–1.044)	0.228
FY2 (N/BW)	0.79 $\pm$ 0.005 (0.070–0.085)	0.801 $\pm$ 0.039 (0.746–0.898)	0.824 $\pm$ 0.025 (0.788–0.869)	0.005
FY3 (N/BW)	0.93 $\pm$ 0.007 (0.79–1.06)	0.767 $\pm$ 0.040 (0.712–0.806)	0.662 $\pm$ 0.049 (0.60–0.712)	0.422
FX1 (N/BW)	-0.052 $\pm$ 0.027 (-0.093–0.016)	-0.061 $\pm$ 0.015 (-0.098– -0.045)	-0.071 $\pm$ 0.012 (-0.098– -0.055)	0.543
FX2 (N/BW)	0.133 $\pm$ 0.030 (0.083–0.180)	0.187 $\pm$ 0.026 (0.109–0.203)	0.224 $\pm$ 0.073 (0.160–0.443)	0.198
FZ (N/BW)	-0.088 $\pm$ 0.058 (-0.175– -0.004)	-0.052 $\pm$ 0.027 (-0.088 – -0.064)	-0.048 $\pm$ 0.013 (-0.069– -0.024)	0.722
Loading rate (BW/S)	0.003 $\pm$ 0.001 (0.002–0.005)	0.002 $\pm$ 0.000 (0.001–0.003)	0.002 $\pm$ 0.000 (0.001–0.002)	0.004
Unloading rate (BW/S)	0.04 $\pm$ 0.04 (0.001–0.02)	0.002 $\pm$ 0.000 (0.002–0.017)	0.004 $\pm$ 0.005 (0.001–0.002)	0.196

p-value adjusted for baseline gait speed; Abbreviation: BW, body weight; CLBP, Chronic low back pain; (SBCLBP): older adults with CLBP and sway-back, (With CLBP): older adults with CLBP, (Without CLBP): older adults without CLBP. CI, Confidence Interval; FY, Vertical force; FZ, Mediolateral force; FX, Anteroposterior force; N, Newton; Values are mean $\pm$ SD

Table 4 The results of asymmetry analysis of Spatiotemporal and force parameters of older adults with SBCLBP, CLBP, and without CLBP participants who participated in the study

Gait parameter	SBCLBP Group (n = 12) Mean $\pm$ SD (min-max)	With CLBP(n = 12) Mean $\pm$ SD (min-max)	Without CLBP(n = 12) Mean $\pm$ SD (min-max)	P-value
Fy1 (%)	0.08 $\pm$ 0.07 (0.00–0.26)	0.07 $\pm$ 0.03 (0.03–0.12)	0.08 $\pm$ 0.04 (0.02–0.15)	0.820
Fy2 (%)	3.96 $\pm$ 4.26 (0.50–15.68)	5.24 $\pm$ 0.83 (4.25–6.62)	3.08 $\pm$ 2.17 (0.38–7.68)	0.181
Fy3 (%)	8.70 $\pm$ 15.29 (0.18–41.40)	2.90 $\pm$ 1.59 (0.05–5.12)	2.75 $\pm$ 1.67 (0.05–5.07)	0.193
Fx1 (%)	-19.64 $\pm$ 20.05 (-69.99 – -3.80)	-22.60 $\pm$ 14.20 (-43.36– -2.91)	-22.70 $\pm$ 14.45 (-42.36– -1.91)	0.885
Fx2 (%)	11.99 $\pm$ 8.65 (1.20–27.42)	6.01 $\pm$ 5.78 (0.41–19.27)	5.93 $\pm$ 5.61 (0.41–18.27)	0.06
Fz (%)	-169.81 $\pm$ 127.04 (-448.52– -12.43)	-237.75 $\pm$ 94.48 (-456.44 – 146.62)	-233.25 $\pm$ 111.59 (-459.35– -105.98)	0.282

Abbreviations: BW, body weight; CLBP, Chronic low back pain; (CLBP and sway back), older adults with CLBP and sway-back; (With CLBP), older adults with CLBP; (Without CLBP), older adults without CLBP; CI, Confidence Interval; FY, vertical force; FZ, mediolateral force; FX, anteroposterior force; N, Newton; %, percentage. Values are mean $\pm$ SD

peak vertical force. The CLBP group showed a higher mean second peak of vertical force than the SBCLBP group, with statistical significance also noted in the loading rates ($p=0.004$). This suggests elevated mechanical stress during gait in participants with sway back and CLBP. The specific values and comparisons are presented in Table 3.

Asymmetry index (ASI)

No significant differences were found in the asymmetry index (ASI) between groups, indicating a similar distribution of load across the limbs among all cohorts. Details of the ASI analysis are provided in Table 4.

Discussion

CLBP is one of several factors impairing sagittal alignment, with aging significantly affecting spinal posture [30]. Over time, these pathologies can contribute to progressive thoracic kyphosis and lumbar lordosis loss. Age-related changes also impact the musculature and ligaments along the spine and lower extremities, particularly in the hip extensors and ankle plantar flexors, which are critical for gait propulsion. This study highlights distinct gait dynamics in older adults with CLBP, including prolonged stance times and reduced gait speeds. These changes likely result from declines in muscular strength, exacerbating postural instability and increasing pain susceptibility [44].

Our findings align with prior research, reinforcing that CLBP significantly impacts mobility in older adults [2, 45]. Changes in posture due to aging influence walking speed [46], a crucial factor in gait analysis. Variations in gait speed affect spatiotemporal parameters, which may explain the reduced ground reaction forces (GRFs) observed in the CLBP group compared to the control group. This could be attributed to decreased hip and knee joint flexion, leading to lower mechanical loads on the legs [13]. Additionally, increased thoracic kyphosis has been linked to reduced hip sagittal mobility, increased external foot progression, and delayed foot-off, potentially explaining the reduced GRFs in elderly individuals with CLBP and sway-back posture [47].

Participants with CLBP adjusted their stride lengths to mitigate impact forces during heel strike, a compensatory mechanism to reduce pain-related discomfort [48–50]. This may also account for the lower second peak vertical force (FY2) observed in older adults with SBCLBP, consistent with findings by Farahpour et al. (2016) on foot pronation's role in modifying GRFs in CLBP patients [51]. In aging individuals, the primary issue may not be overall muscle strength but rather the speed of muscle contraction, which could explain the decreased peak plantar-flexion angle among our older participants [46].

Altered upper torso positioning shifts the body's center of mass toward the plantar aspect of the foot through biomechanical adjustments at the hip and ankle joints. This shift redistributes tension from dorsal musculature to the calf muscles [5]. Additionally, increased trunk stiffness—linked to augmented GRFs—has been observed with reduced quadriceps activity during initial heel contact [52, 53], potentially impairing energy absorption and increasing mechanical stress. Contrary to expectations, the SBCLBP group exhibited the highest loading rates, suggesting that individuals with LBP may adopt compensatory gait strategies that prolong the loading response phase, reducing mechanical efficiency [54].

Our analysis of gait asymmetry revealed that the Asymmetry Index (ASI) did not significantly differ among groups. While previous studies have suggested that CLBP may alter limb load distribution, our findings indicate that individuals with SBCLBP may employ compensatory mechanisms to maintain symmetrical gait patterns [55]. However, setting a threshold for asymmetry would provide a clearer distinction between normal and pathological gait patterns, as minor deviations from symmetry are common and not necessarily indicative of dysfunction. Discrepancies in studies on spinal deformities and scoliosis may stem from variations in frontal plane deformities or the severity of spinal abnormalities [56].

Post hoc analysis was conducted to ensure statistical power calculations were realistic based on our collected data, refining the interpretation of group differences.

Additionally, we acknowledge the challenges in defining sway-back posture objectively. The magnitude of this postural deviation was quantified to ensure that only individuals meeting predefined criteria were included. Future studies should refine these diagnostic criteria to enhance reproducibility and clinical applicability.

Clinical implications

Considering the structural disorders that occur in the body's condition over time, it may be possible to control these changes and prevent the emergence of muscle weaknesses through proper training. A comprehensive understanding of the relationship between body posture defects and the distribution of ground reaction forces on the feet can lead to a more thorough evaluation of how these factors are interconnected. This, in turn, can enhance the effectiveness of treatments for any disorders associated with these phenomena. Continuous gait analysis is essential for monitoring the efficacy of these interventions and for ongoing adjustments to treatment protocols.

Future research should explore the integration of advanced technological aids, such as robotic exoskeletons and virtual reality systems, in rehabilitation settings. These tools offer novel approaches to managing and potentially reversing gait abnormalities associated with chronic back pain, paving the way for improved clinical outcomes and enhanced quality of life for affected individuals.

Limitations

This study provides extensive insights into older adults with SBCLBP, surpassing previous research focused on generic CLBP. However, several limitations should be noted. First, only CLBP patients participated, limiting the generalizability of our findings to other types of LBP. Additionally, while this study focuses on chronic conditions such as LBP and postural changes, other pain conditions may also be relevant in older adults. Future research should explore additional chronic pain conditions, such as joint or muscular pain, to provide a more comprehensive understanding of factors affecting quality of life in this population. Regulating walking speeds in subsequent studies may refine correlations between clinical gait analysis and real-life mobility challenges faced by individuals with CLBP.

Conclusions

Older adults with SBCLBP exhibit significant gait alterations, including slower walking speeds, shorter stride lengths, and lower cadence compared to CLBP and asymptomatic individuals. These individuals also demonstrate prolonged stance phase and stride time. GRF analysis reveals lower FY2 values and higher loading rates in

the SBCLBP group, suggesting that sway-back posture exacerbates the biomechanical deficits associated with CLBP.

Given the limited research on compensatory postures and gait dynamics, further studies are needed to clarify underlying mechanisms and optimize rehabilitative strategies. The use of objective gait assessment tools in clinical practice may aid in identifying gait impairments and improving targeted interventions for this population.

Author contributions

Conceptualization, F.E.; A.L.; M.T.K.; and M.K.; methodology, F.E.; A.L.; M.T.K.; and M.K.; validation, F.E.; A.L.; M.T.K.; and M.K.; investigation, F.E.; A.L.; M.T.K.; and M.K.; data curation, F.E.; A.L.; M.T.K.; and M.K.; writing—original draft preparation, A.L.; G.R.; and J.H.V.; writing—review and editing, A.L.; G.R.; and J.H.V.; visualization, A.L.; supervision, A.L.; G.R.; and J.H.V. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no external funding.

Data availability

Data are available from the corresponding author upon reasonable request.

Declarations

Institutional review board

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethics Committee of the Medical Center Office of Shiraz Rehabilitation Sciences University (IR.SUMS.REHAB.REC.1399.034).

Consent for publication

All participants provided written informed consent for the publication of their personal and clinical details, as well as any identifying images, in this study.

Informed consent

Informed consent was obtained from all subjects involved in the study.

Competing interests

The authors declare no competing interests.

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Received: 18 September 2024 / Accepted: 20 March 2025

Published online: 07 April 2025

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